

ELVIS TATA TANGHANG

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PERSONAL STATEMENT

I am Elvis Tata Tanghang, a Cameroonian-born AI/ML researcher and a graduate of Carnegie Mellon University, where I earned an M.Sc. in Electrical and Computer Engineering. My work centers on speech technology and large language modeling, built on a background in machine learning and deep learning. My main research interest is speech-to-speech translation for low-resource languages, spanning speech recognition (STT), machine translation (MT), text-to-speech synthesis (TTS), and audio language modeling.

EDUCATION

Master of Science in Electrical and Computer Engineering, Advanced Studies: Carnegie Mellon University May 2025, Pittsburgh, PA

Coursework: Machine Learning, Deep Learning, NLP, Generative AI, Speech Technology for Conversational AI.

Bachelor of Electrical and Electronic Engineering: National Higher Polytechnic Institute Dec 2022, Cameroon

Coursework: Software Architecture, Data Structures, Algorithms, Embedded Systems, Engineering Mathematics.

PUBLICATIONS & PREPRINTS

- **Digital Umuganda** “KinyaSpeak: Cache-Based Streaming Automatic Speech Recognition for Kinyarwanda, English, and French.” *Under review, ACL ARR 2026 (May cycle); committed to EMNLP 2026.* First-author work introducing a FastConformer hybrid RNN-T/CTC streaming ASR model (~120M parameters) that matches or surpasses its offline counterpart, documenting a “streaming paradox” for low-resource languages.
- **Aura-ASR: Multilingual Speech Recognition and Translation for African Languages.** *Under review at IEEE Spoken Language Technology Workshop (SLT) 2026.* A CTC-encoder + LLM architecture spanning 20+ African languages, trained on ~18K hours, reaching state-of-the-art WER on several FLEURS, NCHLT, and BembaSpeech benchmarks and surpassing MMS-1B and OmniASR baselines.

SKILLS

Programming	Python (4+ years), C++, SQL, Git
ML frameworks	PyTorch, TensorFlow, scikit-learn, NumPy, Hugging Face, ESPnet, NeMo, TensorBoard, Weights & Biases
Speech & NLP	Multilingual/low-resource ASR; streaming ASR (RNN-T, CTC); speech translation (S2TT, S2ST); machine translation; text-to-speech; audio language modeling; language ID; transfer & few-shot learning
Model development	Multi-stage speech LLMs; quantization; hyperparameter tuning; error analysis; data-efficient training; evaluation
Systems & tools	Jupyter, OpenCV, AWS, Whisper, HPC & multi-GPU training (SLURM, DDP; Leonardo, Bridges-2)
Communication	Technical leadership, cross-functional collaboration, research documentation
Languages	English (native), French (intermediate), Kinyarwanda (beginner), German (basic)

RELEVANT EXPERIENCE

Research Associate: Wakanda AI Fall 2024 – Present
Carnegie Mellon University (CMU Africa) Kigali, Rwanda

- Building **Aura-LLM**: an African-centric, decoder-only large language model trained from scratch to support 25+ African languages. It serves as the base model for downstream research on low-resource African languages, including machine translation, speech recognition/translation, and speech synthesis.
- Built **Aura-ASR**, a multi-stage multilingual speech system (self-supervised CTC encoder → projector → Aura-1B LLM) trained on ~18K hours across 22 African languages, reaching state-of-the-art WER on several FLEURS, NCHLT, and BembaSpeech benchmarks and surpassing MMS-1B and OmniASR baselines.

- **Benchmarking & data.** Contributed to an open benchmarking framework evaluating ASR and speech translation against MMS-1B, SeamlessM4T-v2, and OmniASR on FLEURS, NCHLT, BembaSpeech, BIG-C, and WaxalNLP, and to a unified data index of ~7.66M utterances across 21 African languages (~19,000 hours) standardized at 16 kHz, with synthetic translation targets for Bemba, Hausa, Igbo, and Yoruba.
- **End-to-end speech translation application.** Helped build a real-time speech-to-speech pipeline chaining ASR, MT, and TTS over WebRTC audio transport (LiveKit) with a custom asynchronous orchestrator, demonstrating end-to-end latency under 2 s (~1.7 s Yoruba→English).
- Ran large-scale multi-GPU (DDP) training on the Leonardo and Bridges-2 supercomputers with NeMo and ESPnet; curated and standardized speech datasets for 31 African languages from public sources under data-privacy standards, ensuring diversity across gender, age, and speaking context.
- Model family: huggingface.co/collections/WakandaAI/aura-family-of-llms

Speech AI Researcher: Digital Umuganda Kigali, Rwanda

January 2026 – Present
Kigali-Rwanda

- Developing **KinyaSpeak**, a trilingual (Kinyarwanda, English, French) streaming ASR system: a FastConformer hybrid RNN-T/CTC model (~120M params) with cache-aware chunked attention supporting latencies from 0 ms (real-time captioning) to offline batch decoding.
- Fine-tuned from NVIDIA NeMo on ~6,000 hours of balanced trilingual speech (2,000 h/language from Common Voice, LibriSpeech/MLS, VoxPopuli, and AfriVoice), using language-conditioning tags and BPE tokenizer expansion (1,024 → 4,096) for French and Kinyarwanda orthography.
- Trained on 4× NVIDIA A100 80GB (Leonardo HPC); best model reaches ~19% average WER, with Kinyarwanda streaming (18.1% @ 1040 ms) beating offline (20.7%), revealing a streaming paradox in low-resource languages.
- Proprietary models on Hugging Face: [DigitalUmuganda/Kinyaspeak](#); basis of a first-author paper under review at ACL ARR 2026 / EMNLP 2026.

Software Engineering and Data Science Intern WiredIn Rwanda

June 2024 - Sep 2024
Kigali, Rwanda

- Designed a hybrid recommendation system combining content-based and collaborative filtering (SVD, KNN, vector similarity).
- Integrated NLP-based sentiment analysis of customer feedback, improving recommendation relevance by 27%.
- Built scalable Python data pipelines for user-interaction data and real-time recommendations, with dashboards to communicate model performance to stakeholders.

AI/ML Engineer AI Center of Excellence Africa

May 2024 - Aug 2024
Kenya (Remote)

- Developed computer-vision models for agriculture across Kenya and Rwanda with multilingual interfaces for local-language support.
- Created inference pipelines optimized for low-resource hardware, addressing real infrastructure constraints.
- Built evaluation and testing frameworks to assess performance across diverse African agricultural contexts.

Deep Learning Projects

Fall 2024 - Spring 2025

- Built computer-vision and speech systems in PyTorch, including CNN face recognition with custom loss functions and seq2seq models with CTC loss.
- Optimized neural architectures (ResNet, ConvNeXt, bidirectional LSTM/GRU) for memory efficiency and inference speed while maintaining strong evaluation metrics.
- Applied data augmentation, feature extraction, and variable-length sequence handling to improve generalization.

TEACHING EXPERIENCE

Teaching Assistant: Introduction to Deep Learning (11-785)

Carnegie Mellon University

Fall 2025
Pittsburgh, PA

- Led technical instruction on foundational and advanced architectures (MLPs, CNNs, RNNs, Transformers).
- Guided students through implementing neural systems from scratch, including generative models (VAEs, GANs, diffusion).
- Developed materials for debugging networks, optimizing model performance, and applying advanced training techniques.
- Supported GPU-accelerated training, data-preprocessing pipelines, and model-evaluation strategies. [\[course page\]](#)

Teaching Assistant: Introduction to Deep Learning (11-785)

Carnegie Mellon University

Spring 2025
Pittsburgh, PA

- Ran recitations and office hours on backpropagation, CNNs, sequence models, and attention, supporting a large cohort.
- Mentored students through homework and course projects, from data pipelines to training, tuning, and evaluation.
- Designed and graded assignments, giving detailed feedback on implementation, debugging, and results analysis.

Teaching Assistant: Introduction to Machine Learning

Carnegie Mellon University

[Fall 2025]
Pittsburgh, PA

- Taught core ML foundations: regression, classification, SVMs, decision trees, neural networks, and MLE/MAP estimation.
- Led recitations and office hours, and graded assignments and exams with constructive, individualized feedback.
- Helped students bridge mathematical theory and practical implementation across the course's programming work.

RELEVANT PROJECT WORK

Kinyarwanda Speech Recognition

2024–2025

Independent Research

- Adapted a 580M-parameter w2v-BERT-2.0 Kinyarwanda ASR model to the health domain with LoRA, training 1.86% of the network to reach 5.98% WER with KenLM shallow fusion while preserving general-domain performance.
- Fine-tuned Whisper for Kinyarwanda ASR under data scarcity, with error analysis targeting phonetic and tonal failure modes.
- Released models and live demos: [model](#) · [demo](#)

Multilingual Speech-to-Speech Translation System

Fall 2024

Speech Technology for Conversational AI: Carnegie Mellon University

- Built an end-to-end multilingual S2S system with real-time language identification and routing for English, French, Spanish, and Hindi.
- Implemented an attention-based classifier using a Whisper encoder for language routing, with component switching based on language ID.
- Developed an evaluation framework covering TTS intelligibility, speech quality, ASR WER, latency, and dialog quality.

Computer Vision for African Agricultural Systems

Fall 2024

- Designed CNN-based crop-disease detection optimized for resource-constrained environments.
- Implemented multilingual interface support for local-language instructions and feedback.
- Applied data-augmentation techniques to address limited training data (few-shot learning in practice).

RELEVANT COURSEWORK

Speech Technology for Conversational AI: Speech LLMs, S2ST, S2TT, TTS, ASR, speech diarization, language identification, with practical ESPnet implementation. **Generative AI & Foundation Models:** Transformer architectures, multilingual model development, few-shot and transfer learning for low-resource languages. **Machine Learning & Deep Learning:** Advanced neural architectures, model optimization, evaluation frameworks, responsible AI principles.